

Kamino DVI Solver Benchmark

RSL-RL · 300 iterations · three seeds · mean ± 95% Student-t CI

- Isaac-Ant-Direct: tuned DVI is 4.9× faster than current Kamino and 5.0× faster than PR3570 P-ADMM.
- Isaac-Ant-Direct: tuned DVI remains 2.5× slower than MJWarp and 1.4× slower than PhysX.
- Isaac-Cartpole-Direct: tuned DVI is 2.0× faster than current Kamino and 2.0× faster than PR3570 P-ADMM.
- Isaac-Cartpole-Direct: tuned DVI remains 1.5× slower than MJWarp and 1.0× slower than PhysX.

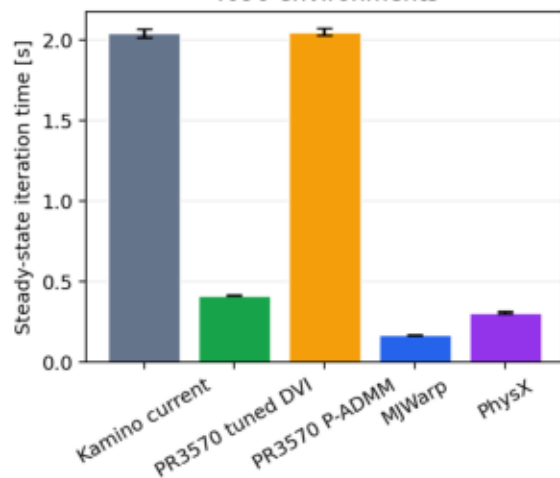
				Total FPS	Reward
Isaac-Ant-Direct	Kamino current	4096	2.040 ± 0.028	64263 ± 890	7402.741 ± 1650.401
Isaac-Ant-Direct	PR3570 tuned DVI	4096	0.413 ± 0.005	317194 ± 3709	10785.473 ± 3745.488
Isaac-Ant-Direct	PR3570 P-ADMM	4096	2.048 ± 0.023	63990 ± 722	8547.088 ± 4094.777
Isaac-Ant-Direct	MJWarp	4096	0.167 ± 0.001	785924 ± 4857	8872.347 ± 33.530
Isaac-Ant-Direct	PhysX	4096	0.304 ± 0.007	433084 ± 9769	6588.875 ± 1341.016
Isaac-Cartpole-Direct	Kamino current	4096	0.261 ± 0.005	251502 ± 4447	4.942 ± 0.005
Isaac-Cartpole-Direct	PR3570 tuned DVI	4096	0.133 ± 0.001	494241 ± 5153	4.950 ± 0.014
Isaac-Cartpole-Direct	PR3570 P-ADMM	4096	0.260 ± 0.002	252178 ± 2015	4.946 ± 0.014
Isaac-Cartpole-Direct	MJWarp	4096	0.087 ± 0.001	749904 ± 9903	4.947 ± 0.009
Isaac-Cartpole-Direct	PhysX	4096	0.133 ± 0.004	494030 ± 12633	-5.542 ± 0.151
Isaac-Velocity-Flat-AnymalD	Kamino current	4096	6.057 ± 0.183	16307 ± 423	21.668 ± 0.335
Isaac-Velocity-Flat-AnymalD	PR3570 tuned DVI	4096	0.889 ± 0.004	110716 ± 450	21.683 ± 0.509
Isaac-Velocity-Flat-AnymalD	PR3570 P-ADMM	4096	6.317 ± 0.107	15606 ± 254	21.722 ± 0.597
Isaac-Velocity-Flat-AnymalD	MJWarp	4096	0.371 ± 0.014	265486 ± 10653	15.859 ± 15.941
Isaac-Velocity-Flat-AnymalD	PhysX	4096	0.479 ± 0.012	205414 ± 5189	19.574 ± 2.938

- Schema v1.1 success series differs from TensorBoard in 30 of 45 runs; report uses TensorBoard success. This is the known generator bug addressed separately in PR 6624.
- Isaac-Ant-Direct tuned DVI success is seed-sensitive; the three-seed 95% CI half-width is 0.539.
- Isaac-Cartpole-Direct PhysX has materially lower final-window reward than current Kamino.

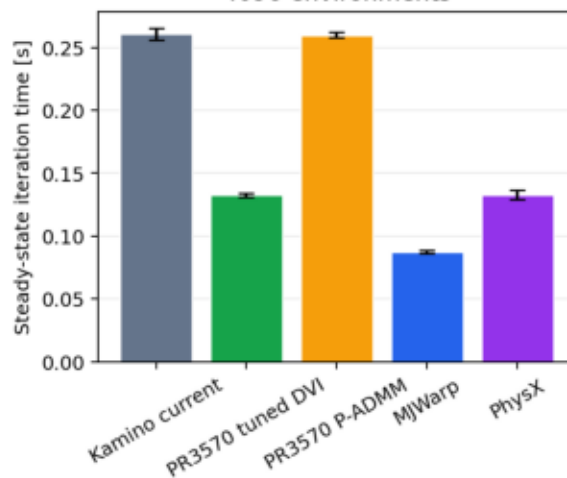
Runtime

RSL-RL runtime (mean $\pm$ 95% CI, three seeds)

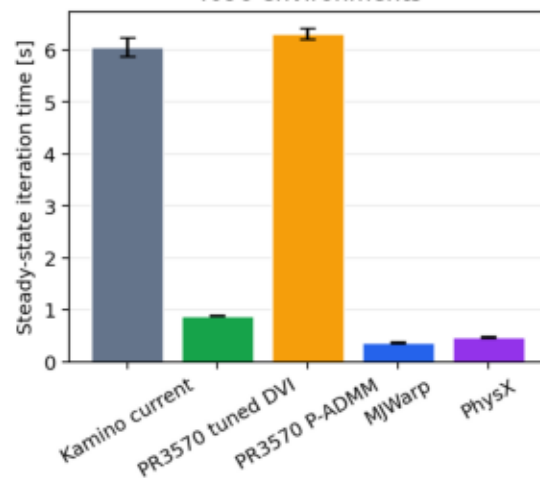
Isaac-Ant-Direct
4096 environments



Isaac-Cartpole-Direct
4096 environments



Isaac-Velocity-Flat-AnymalD
4096 environments



Learning

RSL-RL learning quality (mean $\pm$ 95% CI, three seeds)

